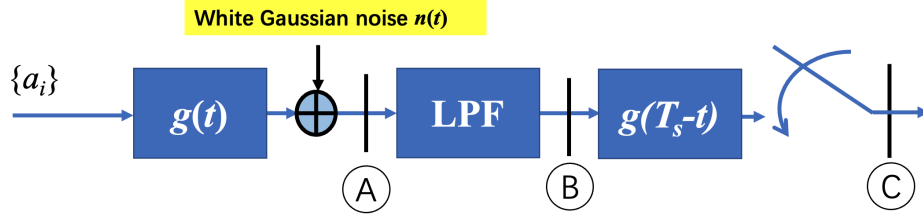


### Solutions to Problem Set 4

1. In this problem, we investigate the SNR (Signal-to-Noise Ratios) at different points in a baseband transmission system with the matched filter and assume no-ISI.



The baseband waveform is a square-root raised cosine with roll-off factor  $\alpha \in [0, 1]$ . At the receiver, the signal is sampled at  $T_s$  after the matched filter. The symbols  $\{a_i\}$  take values in  $\{0, 1, 2, 3\}$  with the equal probability. The double-sided PSD (Power Spectrum Density) of the noise is  $n_0/2$ .

- a) With the no-ISI condition, what is the bandwidth  $W$  of  $G(f)$ ? Write the explicit form of the frequency representation  $G(f)$ .
- b) Derive the SNR at point B, from the baseband waveform perspective. You can keep the integral term of  $G(f)$ .  
(Hint: The distribution of  $\{a_i\}$  should be considered. )
- c) Derive the SNR at point C, after sampling at  $T_s$ . Compare the SNR at point A, B, C and explain.

**Solution.** a) To satisfy the no-ISI condition,

$$W = \frac{1 + \alpha}{2} R_s = \frac{1 + \alpha}{2T_s}$$

The explicit form of the frequency representation  $G(f)$  of the square-root raised cosine is

$$G(f) = \begin{cases} \sqrt{T_s}, & |f| < \frac{1-\alpha}{2T_s} \\ \sqrt{\frac{T_s}{2} \left\{ 1 + \cos \left[ \frac{\pi T_s}{\alpha} \left( |f| - \frac{1-\alpha}{2T_s} \right) \right] \right\}}, & \frac{1-\alpha}{2T_s} \leq |f| \leq \frac{1+\alpha}{2T_s} \\ 0, & |f| > \frac{1+\alpha}{2T_s} \end{cases}$$

b) The SNR at point B is

$$\text{SNR}_B = \frac{E_s R_s}{n_0 W} = \frac{\mathbb{E}[a_i^2] \int_{-\infty}^{\infty} |G(f)|^2 df R_s}{n_0 W}$$

Since  $\mathbb{E}[a_i^2] = 7/2$  and  $W = (\alpha + 1)R_s/2$ , we substitute them and obtain

$$\text{SNR}_B = \frac{7 \int_{-\infty}^{\infty} |G(f)|^2}{(1 + \alpha)n_0}$$

c) The SNR at point A is

$$\text{SNR}_A = \frac{E_s R_s}{n_0 \cdot \infty} = 0$$

The SNR at point C is

$$\text{SNR}_C = \frac{2E_s}{n_0} = 2 \frac{\mathbb{E}[a_i^2] \int_{-\infty}^{\infty} |G(f)|^2 df}{n_0} = \frac{7 \int_{-\infty}^{\infty} |G(f)|^2}{n_0}$$

Finally, we concludes that

$$0 = \text{SNR}_A < \text{SNR}_B \leq \text{SNR}_C$$

At point A, the white Gaussian noise is not band-limited, thus the power is infinity. The SNR at point C is better than point B because the matched filter is coherent with the signal waveform. The equality is reached when  $\alpha = 0$ , i.e. the matched filter is exactly the low-pass filter.

2. In a baseband transmission system, the transmitted signal is

$$x = \begin{cases} 0, & \text{symbol '0' is transmitted} \\ A, & \text{symbol '1' is transmitted} \end{cases},$$

and the received signal is  $y = x + n$ , where  $n$  is the equivalent additive noise, following the Laplace distribution

$$f(n) = \frac{1}{2\lambda} e^{-\frac{|n|}{\lambda}}$$

- a) Assume that the transmitted symbols are equiprobable. Determine the detection threshold with minimum SEP (Symbol Error Probability) and calculate the corresponding SEP.
- b) If the detection rule at the receiver is biased, specifically

$$\tilde{x} = \begin{cases} '0', & y < \frac{1}{4}A \\ '1', & y \geq \frac{1}{4}A \end{cases}.$$

Find the conditional symbol error probability  $P(\tilde{x} = '0'|x = '1')$  and  $P(\tilde{x} = '1'|x = '0')$ .

- c) Find the prior distribution of the transmitted symbols so that the detection rule in b) minimizes the SEP for the given prior distribution.
- (Hint: the question is different from minimizing the SEP given the detection rule)

**Solution.** a) By symmetry, the optimal detection threshold is  $A/2$ . The corresponding SEP is

$$P_e = P(x = 0)P\left(n > \frac{A}{2}\right) + P(x = A)P\left(n < -\frac{A}{2}\right) = \int_{A/2}^{\infty} \frac{1}{2\lambda} e^{-\frac{n}{\lambda}} dn = \frac{1}{2} e^{-\frac{A}{2\lambda}}$$

b) We calculate directly by

$$P(\tilde{x} = '0'|x = '1') = \int_{3A/4}^{\infty} \frac{1}{2\lambda} e^{-\frac{n}{\lambda}} dn = \frac{1}{2} e^{-\frac{3A}{4\lambda}}$$

$$P(\tilde{x} = '1'|x = '0') = \int_{A/4}^{\infty} \frac{1}{2\lambda} e^{-\frac{n}{\lambda}} dn = \frac{1}{2} e^{-\frac{A}{4\lambda}}$$

- c) Since the detection rule in b) is optimal, it satisfies the MAP (Maximum A Posteriori) rule. Therefore the MAP are the same for both hypotheses at the threshold  $A/4$ .

$$P(x = '0')f(A/4) = P(x = '1')f(3A/4)$$

$$\frac{P(x = '0')}{P(x = '1')} = e^{-\frac{A}{2\lambda}}$$

$$P(x = '0') = \frac{e^{-\frac{A}{2\lambda}}}{1 + e^{-\frac{A}{2\lambda}}}, \quad P(x = '1') = \frac{1}{1 + e^{-\frac{A}{2\lambda}}}$$

3. In a baseband transmission system with binary antipodal symbol set, the data rate is 5kbps and the double-sided PSD (Power Spectrum Density) of the additive Gaussian noise is  $n_0/2$ , where  $n_0 = 10^{-9}\text{W/Hz}$ .

- a) If the system is required to be reliable, producing less than 100 bit errors on average in one day, what is the required SEP (Symbol Error Probability) of the system?
- b) Determine the minimum energy per symbol  $E_s$  and the average transmitting power  $P_t$ .  
(Hint: use `qfuncinv(·)` in Matlab to calculate)

**Solution.** a) The require symbol error probability is

$$P_s = \frac{100\text{bits}}{5000\text{kbps} \times 86400\text{s}} = 2.32 \times 10^{-7}$$

b) The error probability of binary antipodal symbols is

$$P_s = Q\left(\sqrt{\frac{2E_s}{n_0}}\right)$$

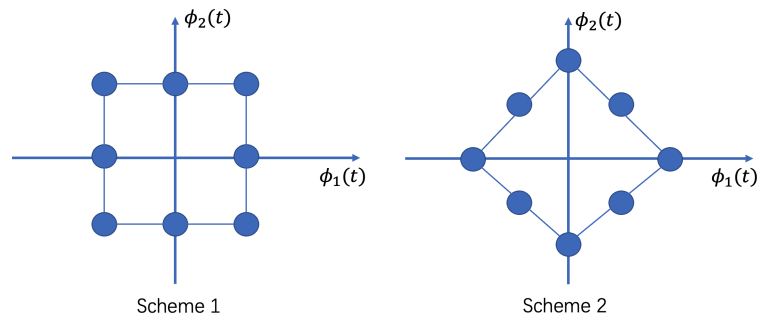
Therefore, the minimum energy per symbol is

$$E_s = \frac{n_0}{2}[Q^{-1}(P_s)]^2 = 1.27 \times 10^{-8}\text{J}$$

The average transmitting power is

$$P_t = E_s R_s = 6.4 \times 10^{-5}\text{W}$$

4. Below are two schemes of 8-symbol constellation. Explain which one is better, in terms of SEP (Symbol Error Probability), given the same power constraint.



**Solution.** The two schemes have exactly the same shape, by rotating  $45^\circ$  you can get the other. Therefore, they have same SEP performance given the same power constraint.

5. In this problem, we compare different methods in passband modulation. Assume the double-sided PSD of noise is  $n_0/2$  and all the symbols are equiprobable.

- a) Given the energy per symbol  $E_s$ , find the SEP (Symbol Error Probability) of 4QAM and 16QAM. Explain why the SEP is lower for 4QAM.
- b) Given the energy per bit  $E_b$ , find the SEP of 4QAM and 16QAM. Which one has a lower SEP?  
(Hint: it is recommended to draw the SEP-SNR curves using Matlab)
- c) Given the energy per bit  $E_b$ , compare the SEP of 16QAM and 16PSK. Explain intuitively why QAM is better?

**Solution.** a) By the SEP of MQAM,

$$P_{s, \text{MQAM}} = 4 \left( 1 - \frac{1}{\sqrt{M}} \right) Q \left( \sqrt{\frac{3E_s}{(M-1)n_0}} \right)$$

For  $M = 4, 16$ , we have

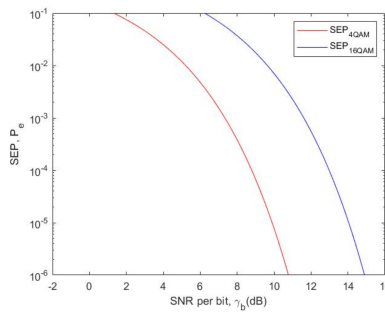
$$P_{s, 4\text{QAM}} = 2Q \left( \sqrt{\frac{E_s}{n_0}} \right) < 3Q \left( \sqrt{\frac{E_s}{5n_0}} \right) = P_{s, 16\text{QAM}}$$

The SEP is lower for 4QAM because for a given average energy per symbol, the constellation points are more separated.

b) Since  $E_b$  is given,  $E_{s, \text{MQAM}} = E_b \cdot \log_2 M$ . Therefore,

$$P_{s, 4\text{QAM}} = 2Q \left( \sqrt{\frac{2E_b}{n_0}} \right) < 3Q \left( \sqrt{\frac{4E_b}{5n_0}} \right) = P_{s, 16\text{QAM}}$$

Still 4QAM has a lower SEP than 16QAM. The SEP-SNR curves is shown below.



c) The SEP of 16PSK is

$$P_{s, 16\text{QAM}} = 2Q \left( \sqrt{\frac{8E_b}{n_0}} \sin \frac{\pi}{16} \right) \approx 2Q \left( \sqrt{\frac{\pi E_b}{32n_0}} \right) > 3Q \left( \sqrt{\frac{4E_b}{5n_0}} \right)$$

where the inequality holds at the typical values of  $E_b/n_0 > 1$ . MQAM is better because it makes full use of the complex plane and thus the minimum distance between the symbols is larger.